14.4 Features of Python over C++

- Python uses Automatic Garbage Collection whereas C++ does not.
- C++ is a statically typed language, while Python is a dynamically typed language.
- Python runs through an interpreter, while C++ is pre-compiled.
- Python code tends to be 5 to 10 times shorter than that written in C++.
- In Python, there is no need to declare types explicitly where as it should be done in C++
- In Python, a function may accept an argument of any type, and return multiple values without any kind of declaration beforehand. Whereas in C++ return statement can return only one value.

Note

Python deletes unwanted objects (built-in types or class instances) automatically to free the memory space. The process by which Python periodically frees and reclaims blocks of memory that no longer are in use is called Garbage Collection.

14.5 Importing C++ Files in Python

Importing C++ program in a Python program is called wrapping up of C++ in Python. Wrapping or creating Python interfaces for C++ programs are done in many ways. The commonly used interfaces are

- Python-C-API (API-**Application Programming Interface** for interfacing with C programs)
- Ctypes (for interfacing with c programs)
- SWIG (Simplified Wrapper Interface Generator- Both C and C++)
- Cython (Cython is both a Python-like language for writing C-extensions)
- Boost. Python (a framework for interfacing Python and C++)
- MinGW (*Minimalist GNU for Windows*)



14.5.1 MinGW Interface

MinGW refers to a set of runtime header files, used in compiling and linking the code of C, C++ and FORTRAN to be run on Windows Operating System.

MinGw-W64 (version of MinGW) is the best compiler for C++ on Windows. To compile and execute the C++ program, you need 'g++' for Windows. MinGW allows to compile and execute C++ program dynamically through Python program using g++.